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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING (24CI3201)
A.Y. 2026 - 2027

<table>
<tr><td>TITLE OF THE PROJECT</td><td colspan="2">Smart Restaurant Management System</td></tr>
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Abstract

Smart Restaurant Management System

The Smart Restaurant Management System is a web-based application designed to simplify and automate the day-to-day operations of a restaurant. Traditional restaurant management often involves manual order handling, billing, inventory tracking, and communication between customers, waiters, chefs, and owners, which can lead to delays and errors. This project aims to provide a digital solution that improves operational efficiency, enhances customer satisfaction, and supports better business management through modern software engineering practices.

The system allows customers to view digital menus and place orders, while waiters can manage table orders efficiently. Chefs receive real-time order updates, enabling faster food preparation, and restaurant owners can monitor sales, inventory, and business performance through an interactive dashboard. The application is developed using the Agile Software Development Life Cycle, where requirements are managed as user stories, development is carried out in iterative sprints, and project management tools such as Jira or Trello support planning, collaboration, and progress tracking.

To ensure reliable software delivery, the project incorporates DevOps practices such as Git-based version control, Continuous Integration (CI), Docker containerization, Kubernetes orchestration, and YAML-based deployment configuration. These technologies enable consistent application deployment, easier maintenance, and improved scalability across different environments.

Security is strengthened through the implementation of SecOps practices. Static code analysis is performed using SonarQube, vulnerability assessment is carried out using OWASP ZAP, and Docker container images are scanned with Trivy to identify security risks before deployment. Secure coding practices, input validation, authentication mechanisms, and OWASP Top 10 guidelines are followed to build a secure and reliable application.

The project also introduces basic MLOps concepts by utilizing restaurant data for future business intelligence and predictive analytics. Machine learning workflows, including data collection, model versioning, deployment using MLflow, and performance monitoring with Prometheus, can be integrated to support features such as demand forecasting, menu recommendations, and sales trend analysis. The application

is designed for cloud deployment using platforms such as AWS, Google Cloud Platform (GCP), or Microsoft Azure, ensuring scalability and easy access.

Overall, the Smart Restaurant Management System demonstrates the practical integration of Agile, DevOps, SecOps, and MLOps into a real-world application. The project not only improves restaurant operations through automation but also showcases modern software development practices, secure deployment strategies, and cloud-based technologies, making it an effective and industry-relevant solution for digital restaurant management.

Signature of the Guide HOD